

AI_Kyt_Sr

1. List and explain six disciplines that compose most of AI.

- Natural Language Processing: to enable it to communicate successfully.
- Knowledge representation: to store what it knows or hears.
- Automated reasoning: to use the stored information to answer questions and to draw

new conclusions.

- Machine Learning: to adapt to new circumstances and to detect and extrapolate patterns.
- Computer Vision: to perceive objects
- Robotics: to manipulate objects and move about.

2. Discuss the Four categories of AI organized.

-Systems that think like humans: The exciting new effort to make computers think.

-Systems that think rationally: The study of mental faculties through the use of computational models

-Systems that act like humans: The art of creating machines that perform functions that require intelligence when performed by people.

-Systems that act rationally: Computational intelligence is the study of the design of intelligent agents.

3. To become a rational agent which facts are required? (OR) What is rational at any given time?

What is rational at any given time depends on four things:

1. The performance measure that defines the criterion of success.
2. The agent's prior knowledge of the environment.
3. The actions that the agent can perform.

4. The agent's percept sequence to date.

4. How can an algorithm's performances evaluated?

Evaluate an algorithm's performance in four ways:

Completeness: Is the algorithm guaranteed to find a solution when there is one?

Optimality: Does the strategy find the optimal solution?

Time Complexity: How long does it take to find a solution?

Space Complexity: How much memory is needed to perform the search?

5. Explain Simple reflex agent with diagram.

-selects actions on the basic of the current percept , ignoring the rest of the percept history.

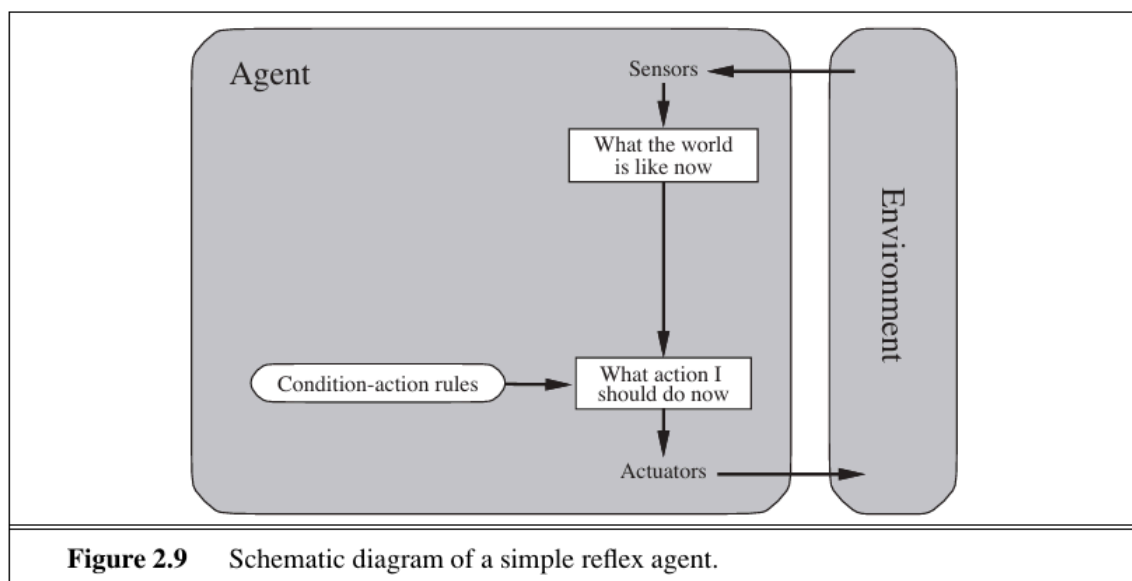
-it uses condition-action rules

-it has admirable property of being simple but very limited intelligence.

-work only if the correct decision can be made on the basic of only the current percept.

work only if the environment is fully observable.

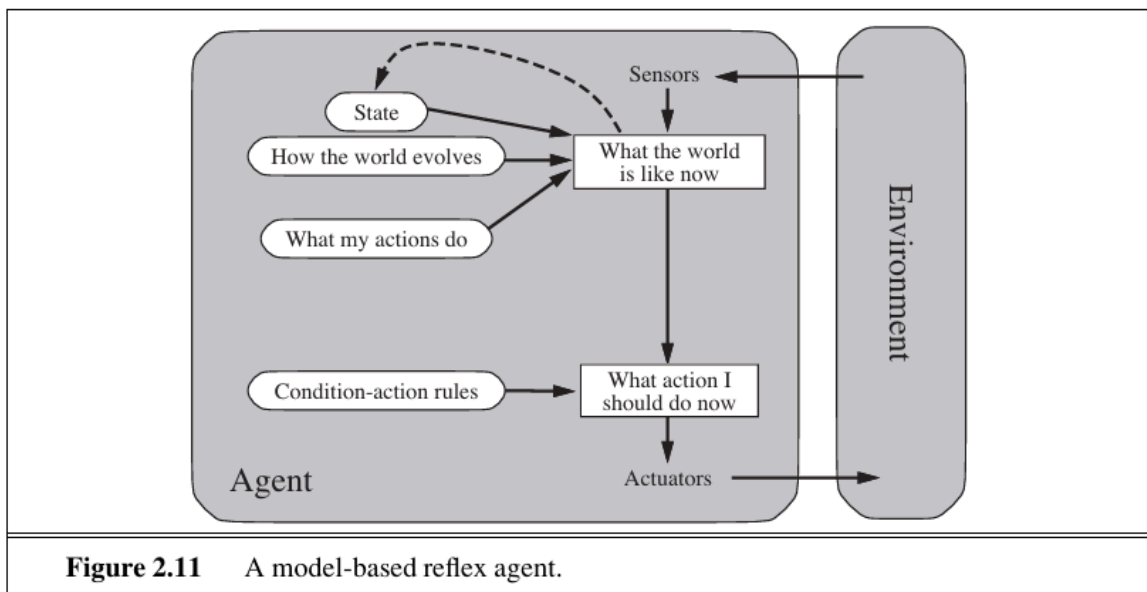
Figure 2.9



6. Explain Model-based reflex agents with diagram.

- should maintain some sort of internal state that depends on the percept history.
- reflects at least some of the unobserved aspects of the current state.
- needs to keep track of the current state of the world using an internal model
- requires two kinds of knowledge to be encoded in the agent program
 - need some information about how the world evolves independently of the agent
 - need some information about how the agent's own actions affect the world

Figure 2.11

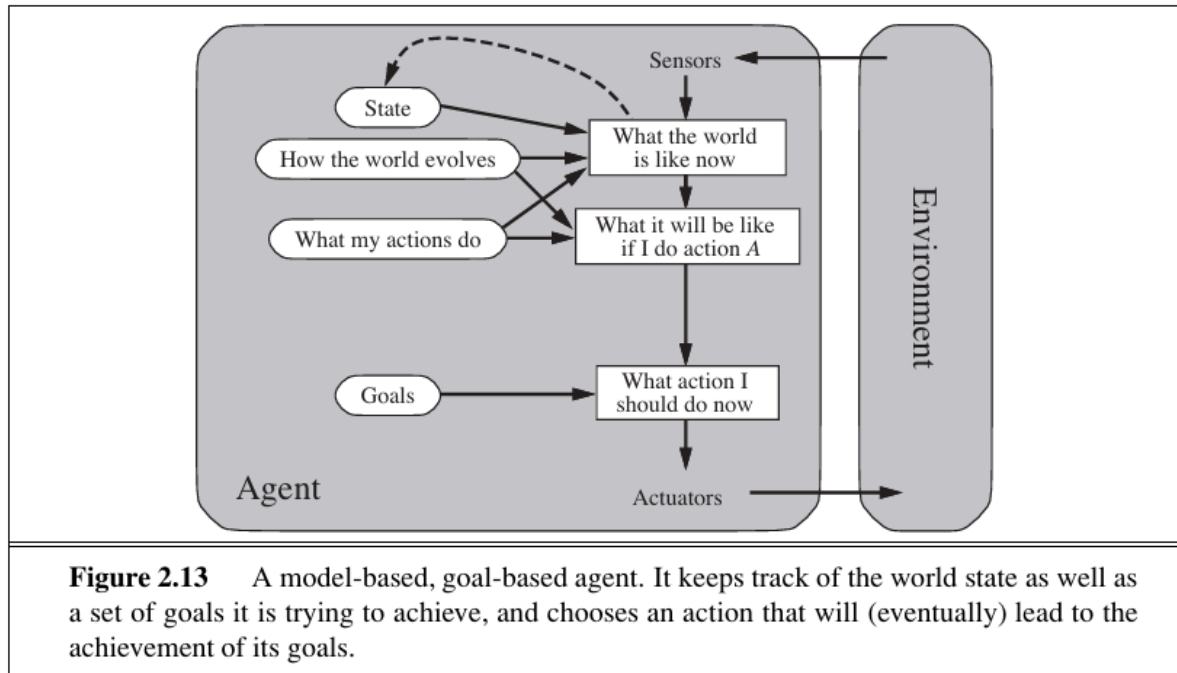


7. Explain Goal-based agents with diagram.

- can combine with goal information about the results of possible actions in order to choose actions that achieve the goal.
- goals are achieved by Search and planning (twists and turns to find a way to achieve the goal)
- less efficient

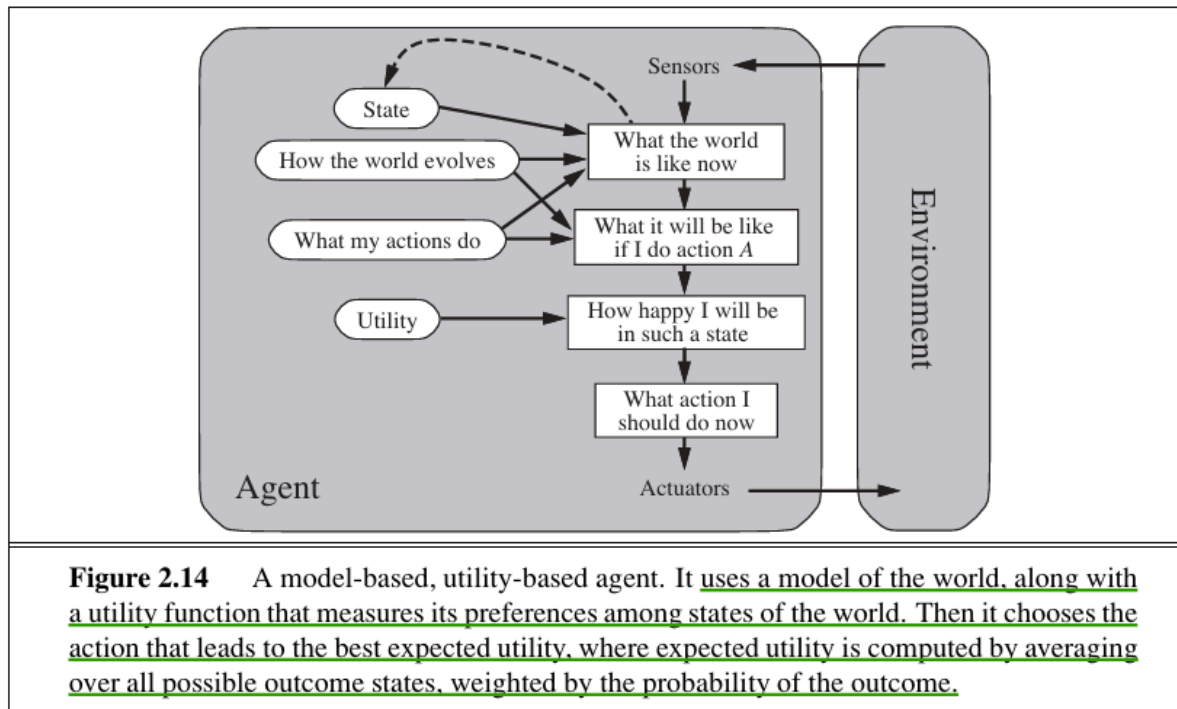
-it is more flexible than model-based reflex agents.

Figure 2.13



8. Explain Utility-based agents with diagram.

- provide a crude binary distinction between "happy" and "unhappy" states.
- maps a state (or a sequence of states) onto a real number, which describes the associated degree of happiness.
- allows rational decisions in two kinds of cases where goals are inadequate.
 - conflicting goals , only some of which can be achieved.
 - several goals that the agent can aim for
- provides a way in which the likelihood of success can be weighed up against the importance of the goals.



9. Explaining Learning agent with diagram.

The agent to operate in initially unknown environments, and to become more competent than its initial knowledge alone might allow.

A learning agent can be divided into four conceptual components. The most important distinction is between the learning element, which is responsible for making improvements and the performance element, which is responsible for selecting external actions.

It takes in percepts and decides on the actions. The learning element uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.

The critic tells the learning element how well the agent is doing with respect to a fixed performance standard.

The last component is the problem generator. It is responsible for suggesting actions that will lead to new and informative experiences. The problem generator's job is to suggest these exploratory actions.

Figure 2.15

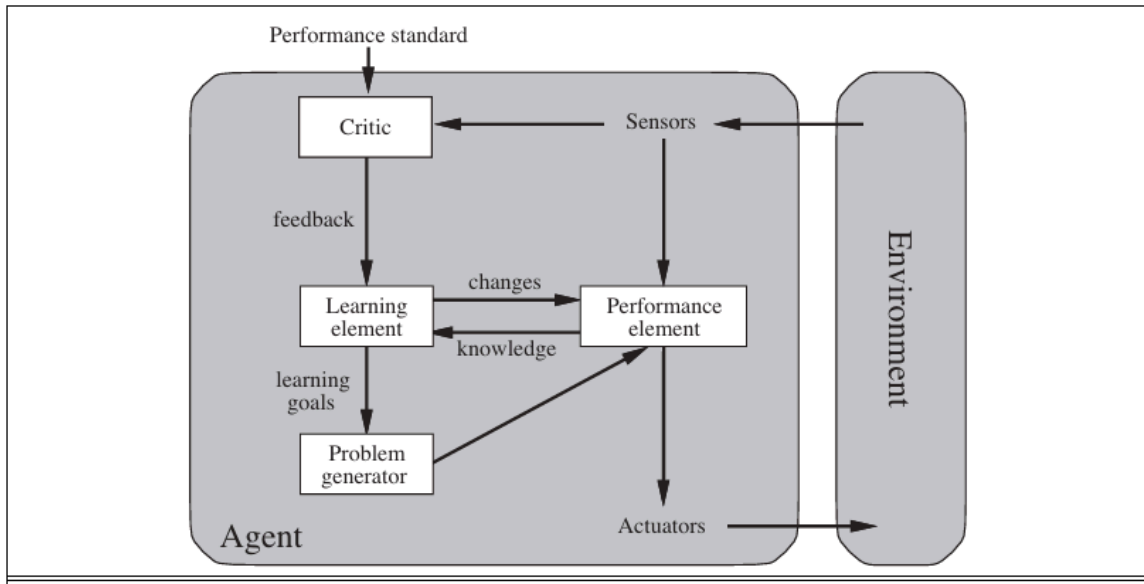


Figure 2.15 A general learning agent.

10. Explain A* search.

-it evaluates nodes by combining $g(n)$, the cost to reach the node, and $h(n)$, the cost to get from the node to the goal.

-it is both complete and optimal

-the space complexity is still prohibitive

$$f(n) = g(n) + h(n)$$

$g(n)$ = path cost from start node to node n

$h(n)$ = estimated cost of the cheapest path from node n to a goal node

$f(n)$ = estimated cost of the cheapest solution through n

11. What are the differences between Blind search and Heuristic search?

Uniformed search/Blind search

-uninformed search strategies have no additional information about states beyond that provide in the problem definition.

-it can generate successors and distinguish a goal state from a nongoal state.

-it is called blind search.

Informed search/Heuristic search

-informed search strategies use problem-specific knowledge beyond the definition of the problem itself

-it can find solution more efficiently than uninformed search strategies.

-it is also called heuristic search

12. Explain three concepts that apply to all forms of propositional logic?

Three concepts that apply to all forms of logic are logical equivalence, validity and satisfiability.

-**Logical equivalence** means that two sentences α and β are logically equivalent if they are true in the same set of models.

-A sentence is **valid** if it is true in all models. Valid sentences are also known as tautologies.

-A sentence is satisfiable if it is true in some model.

13. Discuss a comparison between declarative approach and procedural approach?

Declarative approach

Designing the representation language to make it easy to express this knowledge in the form of sentence simplifies the construction problem enormously. This is called declarative approach to system building.

Procedural approach

Procedural approach encodes desired behaviors directly as program code, minimizing the role of explicit representation and reasoning can result in much more efficient system.

14. What are the four components of a Problem?

A problem can be defined formally by four components. The four components of a problem are initial state, successor function, goal test and path cost.

-The **initial state** that the agent starts in.

-a description of the possible actions available to the agent. The most common formulation uses a **successor function**. The initial state and successor function implicitly define the state space of the problem.

- The goal test, which determines whether a given state is a goal state . Sometimes there is an explicit set of possible goal states, and the test simply checks whether the given state is one of them.
- A path cost function that assigns a numeric cost to each path. The problem-solving agent choose a cost function that reflects its own performance measure.

15. **Difference between propositional logic and first-order-logic.(Chapter 8)**

Propositional Logic

- Propositional logic assumes the world contains facts
- it is a declarative language
- it is compositional
- it has very limited expressive power

First-order-Logic

- First-order logic assumes the world contains objects,relations and functions
- it is an extensions of propositional logic, so it is also declarative language.
- it is sufficiently expressive to represent the natural language statements in a concise way.

16. **What are the basic syntactic elements of the first-order-logic?(Chapter-8)**

The basic syntactic elements of first-order logic are the symbols that stand for objects, relations and functions.

There are three types of symbols:

- (1) constant symbols which stand for objects
- (2) predicate symbols which stand for relations
- (3) function symbols which stand for functions

17. **Explain the concept of a Bayesian Network and describe its main components and applications. Chapter 7**

`Bayesian Networks utilize knowledge about the independence of variables to simplify the model.

The main components of a Bayesian Network are:

- Nodes: Represent variables or events , such as Rain,Cloudy or Traffic.
- Directed edges; Represent dependencies between variables.
- Conditional Probability Tables(CPTs): Specify the probability of each variable given the values of its parent variables.
- Directed Acyclic Graph: Ensures that the network does not contain any directed cycles.

18. What is a Conditional Probability Table(CPT)?Explain its purpose and role in Bayesian Network?

Each node in the network represents a variable and every directed edge a statement of conditional probability. The CPT (conditional probability table) of a node lists all the conditional probabilities of the node's variable conditioned on all the nodes connected by incoming edges. The two nodes Bur and Ear are not linked since the variables are independent. All other nodes have a parent node , which makes the reasoning a little more complex.